

Istanbul BFT konsenzus algoritam

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Model sistema

- deterministički konsenzus algoritam zasnovan na lideru i delimično otporan na vizantijske greške ($n \geq 3f + 1$)
- delimično sinhron komunikacioni model
- tokom perioda sinhronosti:
 - latencija = 3 (završetak odlučivanja za 3 kašnjenja poruka)
 - komunikaciona složenost $O(n^2)$
- sistem se sastoji od poznatog skupa n procesa: $\Pi = \{p_1, p_2, \dots, p_n\}$
- Δ - gornja granica za vreme izvršavanja i kašnjenja komunikacije
- GST (global stabilization time)
- pp: svaki proces ima pristup broadcast aplikaciji koja šalje poruku svakom procesu iz skupa procesa, uključujući i samom sebi
- osobine: *agreement, validity i termination*

Pseudokod 1/4

- pseudokod prikazuje kako se 1 instanca konsenzusa izvršava na jednom ispravnom procesu
- Algoritam 1 sadrži konstante, promenljive stanja i proceduru Start koja pokreće instancu konsenzusa na procesu

Algorithm 1 IBFT pseudocode for process p_i : constants, state variables, and ancillary procedures

```
1: constants:  
2:    $p_i$  ▷ The identifier of the process  
  
3: state variables:  
4:    $\lambda_i$  ▷ The identifier of the consensus instance  
5:    $r_i$  ▷ The current round  
6:    $pr_i$  ▷ The round at which the process has prepared  
7:    $pv_i$  ▷ The value for which the process has prepared  
8:    $inputValue_i$  ▷ The value passed as input to this instance  
  
9: timer:  
10:   $timer_i$   
  
11: procedure START( $\lambda$ ,  $value$ )  
12:   $\lambda_i \leftarrow \lambda$   
13:   $r_i \leftarrow 1$   
14:   $pr_i \leftarrow \perp$   
15:   $pv_i \leftarrow \perp$   
16:   $inputValue_i \leftarrow value$   
17:  if LEADER( $h_i$ ,  $r_i$ ) =  $p_i$  then  
18:    broadcast  $\langle \text{PRE-PREPARE}, \lambda_i, r_i, inputValue_i \rangle$  message  
19:    set  $timer_i$  to running and expire after  $t(r_i)$ 
```

Pseudokod 2/4

- Algoritam 2 je normal case operation, pokriva situaciju kada je lider ispravan, poruke stižu na vreme i nema potrebe za promenom runde
- PRE-PREPARE -> PREPARE -> COMMIT -> DECIDE

Algorithm 2 IBFT pseudocode for process p_i : normal case operation

```
1: upon receiving a valid  $\langle \text{PRE-PREPARE}, \lambda_i, r_i, \text{value} \rangle$  message  $m$  from  $\text{LEADER}(\lambda_i, \text{round})$  such that  $\text{JUSTIFYPREPREPARE}(m)$  do  
2:   set  $\text{timer}_i$  to running and expire after  $t(r_i)$   
3:   broadcast  $\langle \text{PREPARE}, \lambda_i, r_i, \text{value} \rangle$   
  
4: upon receiving a quorum of valid  $\langle \text{PREPARE}, \lambda_i, r_i, \text{value} \rangle$  messages do  
5:    $pr_i \leftarrow r_i$   
6:    $pv_i \leftarrow \text{value}$   
7:   broadcast  $\langle \text{COMMIT}, \lambda_i, r_i, \text{value} \rangle$   
  
8: upon receiving a quorum  $Q_{\text{commit}}$  of valid  $\langle \text{COMMIT}, \lambda_i, \text{round}, \text{value} \rangle$  messages do  
9:   set  $\text{timer}_i$  to stopped  
10:  DECIDE( $\lambda_i, \text{value}, Q_{\text{commit}}$ )
```

Pseudokod 3/4

- Algorithm 3 služi za liveness, jer omogućava algoritmu da nastavi rad kada trenutna runda ne uspe

Algorithm 3 IBFT pseudocode for process p_i : round changes

```
1: upon  $timer_i$  is expired do
2:    $r_i \leftarrow r_i + 1$ 
3:   set  $timer_i$  to running and expire after  $t(r_i)$ 
4:   broadcast  $\langle \text{ROUND-CHANGE}, \lambda_i, r_i, pr_i, pv_i \rangle$ 

5: upon receiving a set  $F_{rc}$  of  $f + 1$  valid  $\langle \text{ROUND-CHANGE}, \lambda_i, r_j, -, - \rangle$  messages such
   that  $\forall \langle \text{ROUND-CHANGE}, \lambda_i, r_j, -, - \rangle \in F_{rc} : r_j > r_i$  do
6:   let  $\langle \text{ROUND-CHANGE}, h_i, r_{min}, -, - \rangle \in F_{rc}$  such that:
7:      $\forall \langle \text{ROUND-CHANGE}, \lambda_i, r_j, -, - \rangle \in F_{rc} : r_{min} \leq r_j$ 
8:    $r_i \leftarrow r_{min}$ 
9:   set  $timer_i$  to running and expire after  $t(r_i)$ 
10:  broadcast  $\langle \text{ROUND-CHANGE}, \lambda_i, r_i, pr_i, pv_i \rangle$ 

11: upon receiving a quorum  $Q_{rc}$  of valid  $\langle \text{ROUND-CHANGE}, \lambda_i, r_i, -, - \rangle$  messages such
    that  $\text{LEADER}(\lambda_i, r_i) = p_i \wedge \text{JUSTIFYROUNDCHANGE}(Q_{rc})$  do
12:   if  $\text{HIGHESTPREPARED}(Q_{rc}) \neq \perp$  then
13:     let  $v$  such that  $(-, v) = \text{HIGHESTPREPARED}(Q_{rc})$ 
14:   else
15:     let  $v$  such that  $v = \text{inputValue}_i$ 
16:   broadcast  $\langle \text{PRE-PREPARE}, \lambda_i, r_i, v \rangle$ 

    ▷ We can omit this if we assume some mechanism external to the consensus algorithm that ensures
    synchronization of decided values.

17: upon receiving a valid  $\langle \text{ROUND-CHANGE}, \lambda_i, -, -, - \rangle$  message from  $p_j \wedge p_i$  has decided
    by calling  $\text{DECIDE}(\lambda_i, -, Q_{commit})$  do
18:   send  $Q_{commit}$  to process  $p_j$ 
```

Pseudokod 4/4

- safety mehanizam kod promene runde

Algorithm 4 IBFT pseudocode for process p_i : message justification

```
1: predicate JUSTIFYROUNDCHANGE( $Q_{rc}$ )
2:   return
      $\forall \langle \text{ROUND-CHANGE}, \lambda_i, r_i, pr_j, pv_j \rangle \in Q_{rc} : pr_j = \perp \wedge pv_j = \perp$ 
      $\vee$  received a quorum of valid  $\langle \text{PREPARE}, \lambda_i, pr, pv \rangle$  messages such that:
          $(pr, pv) = \text{HIGHESTPREPARED}(Q_{rc})$ 

3: predicate JUSTIFYPREPREPARE( $\langle \text{PRE-PREPARE}, \lambda_i, round, value \rangle$ )
4:   return
      $round = 1$ 
      $\vee$  received a quorum  $Q_{rc}$  of valid  $\langle \text{ROUND-CHANGE}, \lambda_i, round, pr_j, pv_j \rangle$  messages
     such that:
          $\forall \langle \text{ROUND-CHANGE}, \lambda_i, round, pr_j, pv_j \rangle \in Q_{rc} : pr_j = \perp \wedge pr_j = \perp$ 
          $\vee$  received a quorum of valid  $\langle \text{PREPARE}, \lambda_i, pr, value \rangle$  messages such that:
              $(pr, value) = \text{HIGHESTPREPARED}(Q_{rc})$ 

 $\triangleright$  Helper function that returns a tuple  $(pr, pv)$  where  $pr$  and  $pv$  are, respectively, the prepared round
and the prepared value of the ROUND-CHANGE message in  $Q_{rc}$  with the highest prepared round

5: function HIGHESTPREPARED( $Q_{rc}$ )
6:   return  $(pr, pv)$  such that:
      $\exists \langle \text{ROUND-CHANGE}, \lambda_i, round, pr, pv \rangle \in Q_{rc} :$ 
      $\forall \langle \text{ROUND-CHANGE}, \lambda_i, round, pr_j, pv_j \rangle \in Q_{rc} : pr_j = \perp \vee pr \geq pr_j$ 
```

Agreement

- Ako jedan ispravan proces odluči vrednost v , nijedan drugi proces ne može odlučiti neku drugu vrednost v'
- Lema 1: Ako je neki ispravan proces postao prepared za vrednost v u rundi r , onda nijedan drugi ispravan proces ne može postati prepared za neku drugu vrednost v' u istoj toj rundi.
- Lema 2: Ako je neka vrednost v već bila prepared kod dovoljno ispravnih procesa u nekoj rundi r , onda u kasnijoj rundi ne sme biti opravdano predložiti neku drugu vrednost v' .
- Agreement se dokazuje tako što se prvo pokaže da konflikt ne može nastati u istoj rundi, a zatim da ne može nastati ni pri promeni runde.

Validity i Termination

- Ako ispravan proces odluči vrednost v , tada mora važiti $\beta(v) = \text{true}$
- Predikat β predstavlja spoljašnju proveru validnosti vrednosti
- Validity se u pseudokodu vidi kroz pojam valid message, jer proces prihvata samo validne poruke
- Termination znači da će svaki ispravan proces na kraju doneti odluku.
- Ideja dokaza:
 - Ako algoritam ne napreduje, tajmer ističe
 - Procesi tada prelaze u novu rundu pomoću ROUND-CHANGE poruka

HVALA NA PAŽNJI